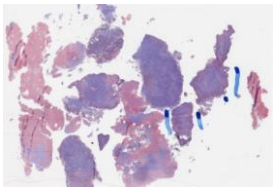
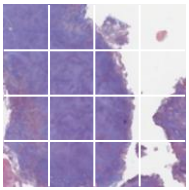
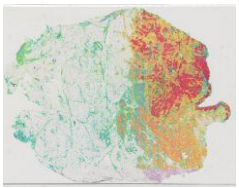
The audited model



Whole-slide image (H&E)

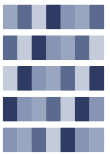


Cut into tiles (256 × 256 px)



CONCH labels each tile

check: how decisively?



UNI-2: one vector per tile

8 compartments in bladder (BLCA), 9 in breast (BRCA); each tile belongs to one

One encoder per compartment

- Tumour tiles → encoder → h_1
- Necrosis tiles → encoder → h_2
- Adipose tiles → encoder → h_3
- … every compartment → encoder → h_R

Cross-region attention

One learned weight per compartment

w_1, w_2, w_3, w_R

Σ

Risk score

Observed outcome (progression-free interval)

check first: performance against clinical-only Cox, ABMIL and Kaplan–Meier

weights

risk score

risk vs outcome

I Interpretability

Read the model's own weights

WHAT IT SEEMS TO SAY

"This tissue drives the risk."

HOW THE AUDIT CHECKS IT

1. Hide compartments in every combination (2^8 or 2^9 sets).
2. Score each set twice: each patient's risk score, and the C-index against outcome. (risk, C-index)
3. Compare both with the weights; repeat on unseen hospitals and on a planted signal.

PASSES IF

Weights order the tissues as the model uses them, and that use orders them as outcome does.

E Explainability

Explain the score from outside

WHAT IT SEEMS TO SAY

"These features explain the risk."

HOW THE AUDIT CHECKS IT

1. A random forest learns the risk score from clinical and pathomic features.
2. Rank the features by importance.
3. Re-rank with a second estimator; test each feature against observed outcome. ($q < 0.05$?)

PASSES IF

The ranking survives a change of estimator, and features survive multiple-testing control against outcome.

R Reliability

Find where the model errs

WHAT IT SEEMS TO SAY

"The model is more or less reliable for these patients."

HOW THE AUDIT CHECKS IT

1. Error per patient over comparable pairs, aware of censoring.
2. MED3pa learns rules that flag groups, on discovery patients. (stage = I ?)
3. On held-out patients: C-index inside each group; 90% bounds per subgroup.

PASSES IF

A flagged group's C-index interval excludes the all-patient value (and bounds hold where promised).

Each explanation is checked against the model and against what happened to patients.